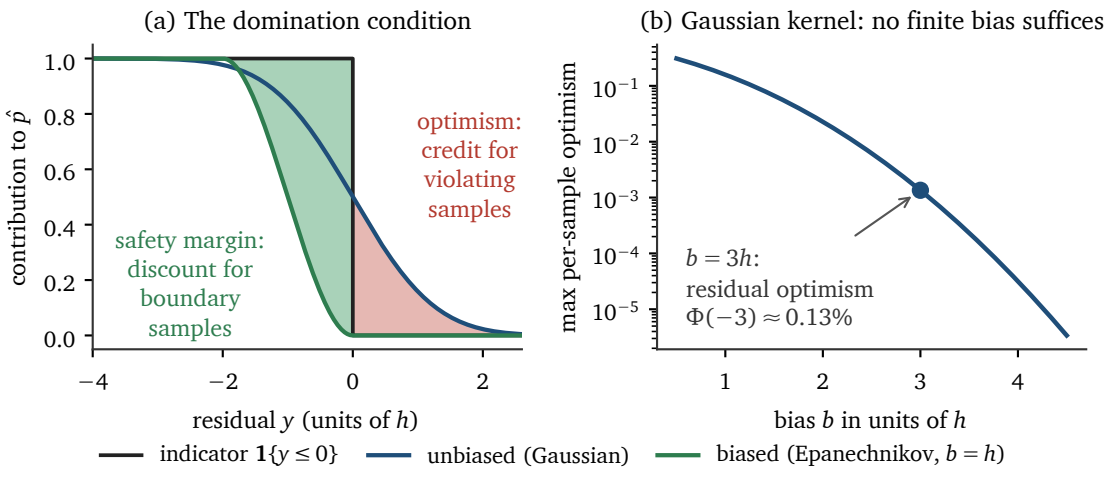
(a) The domination condition
contribution to $\hat{p}$
1.0
0.8
0.6
0.4
0.2
0.0
optimism:
credit for
violating
samples
safety margin:
discount for
boundary
samples
−4
−2
0
2
residual $y$ (units of $h$)
(b) Gaussian kernel: no finite bias suffices
max per-sample optimism
$10^{-1}$
$10^{-2}$
$10^{-3}$
$10^{-4}$
$10^{-5}$
$b = 3h$:
residual optimism
$\Phi(-3) \approx 0.13\%$
1
2
3
4
bias $b$ in units of $h$
indicator $\mathbf{1}\{y \le 0\}$
unbiased (Gaussian)
biased (Epanechnikov, $b = h$)